

Javier Chang (常中嶺)

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Skills

- **Firmware:** C, Arduino, UART / Serial Protocols
- **Backend:** .NET (C# / .NET Core), PHP (Laravel), Python (Django), C
- **Frontend & Mobile:** Vue.js (RWD), Flutter, Android (Java), UWP
- **Database & DevOps:** MSSQL, MySQL, Enterprise Virtualization (vSphere / Horizon VDI)

Patent

- M455937 - Intelligent LED-Guided Directional Moving Floor Escape System
- M487500 - Smart Escape Route Guidance and Directional Indicator System
- M476895 - Multifunctional lamps
- M483873 - Instant Scan-and-Edit Smart Printer with Wireless Transfer

Work Experience

Institute for Information Industry (III) | Full-Stack Engineer

2023.12 - Present

- Led system analysis (SA), technical documentation, and MSSQL schema design for the "New Performance Management System"; systematically resolved cross-departmental edge cases, reducing routine communication and maintenance overhead between IT and HR by 40%.
- Implemented a decoupled MVC architecture using .NET Core and Vue.js; designed high-concurrency Stored Procedures for large-scale data matching, replication, and core CRUD business logic, establishing rigorous validation to ensure data consistency.
- Developed an end-to-end online approval workflow module (including dynamic nodes for draft, submission, rejection, withdrawal, and approval), fully digitizing traditional paper-based processes to eliminate delays and archival inefficiencies.
- Managed legacy system maintenance, code refactoring, and comprehensive QA testing; successfully diagnosed logical flaws in legacy performance and manager-evaluation subsystems, leveraging these insights to ensure a seamless upgrade with zero major defects upon launch.

Freelancer | Full-Stack Engineer (Side Projects)

2023.12 - 2024.07

- Designed modern GUIs and implemented low-level hardware control logic using Python; leveraged the UART serial communication protocol to achieve low-latency, real-time command transmission and bidirectional data feedback.
- Independently developed an "Online English Certification Website" utilizing a decoupled MVC architecture with Vue, Laravel, and MySQL; integrated an automated grading mechanism and a dynamic course reservation system to streamline and automate teaching workflows.
- Built a customized official website for a travel agency, leveraging Vue and the Google Drive API ecosystem to replace legacy systems; upgraded data and payment security protocols, slashing the client's maintenance and operational costs by over 80%.

China Broadcasting Corporation (BCC) | Full-Stack Engineer

2021.10 - 2023.11

- Maintained and optimized large-scale .NET Framework & MSSQL systems; integrated third-party payment gateways, automated electronic invoicing, and data encryption verification. Implemented automated background tasks via .NET and enforced cybersecurity controls using VPNs and firewalls.
- Participated in the preparation of the new e-commerce website, leading system analysis (SA) and requirements gathering; defined core business logic, technical specifications, and database migration strategies to ensure a seamless system transition and architectural upgrade.

Educational

National Taiwan University > M.S. in CSIE | GPA: 3.87

- Journal IEEE - Improving the Accuracy Rate of PM2.5 Pollution Forecast Using Q-learning training Time-series Model (IEEE INSPEC Accession Number: 19313037)

National Taiwan University of Science and Technology > B.S. in CSIE | GPA: 3.97 (Ranking: 2/52)

- Intelligent LED-Guided Directional Moving Floor Escape System (C, ARM)
- First Place – National LED Creative Design Competition
- Excellence Award – TiC100 Smart City & IoT Business Model Competition
- First Place – CTCL Foundation Science and Technology Scholarship

Taiwan Data Science Co., Ltd | Development Engineer

2019.02 - 2021.09

(Hardware Control & IoT Security)

- Real-time Payment & Passenger Flow Monitoring System (SA / Architecture Design): Led system and AI architecture planning to shift defenses forward via "pre-payment real-time blocking." Leveraged DDPG reinforcement learning and data fusion to overcome edge log scarcity, incorporating antenna spatial features to trigger instant transaction circuit-breaking. Implemented dual-track forensics using HMM and LSTM-CNN to analyze anomalous sequences and deep packet timing for attack trajectory reconstruction; integrated Policy Gradient and Q-learning for per-log online dynamic learning, enabling self-evolving system capabilities.
- Taipei Medical University - Mobile Medical App & Hardware Integration (COPD Analysis Project - Sole Developer): Developed a mobile medical application using Flutter; deeply integrated C/Arduino firmware via BLE (Bluetooth Low Energy) to achieve low-latency, bidirectional, real-time transmission of medical parameters.
- Commercial Automated Control System: Independently engineered the full-stack system using UWP (.NET Framework); spearheaded UI design and data logic while implementing low-level hardware bidirectional command control through APIs.

(Full-Stack & Systems)

- **【Wei-Mei-You】**Nutrition Management CRM (Sole Developer): Spearheaded UI design, frontend-backend integration, and database schema using .NET Core, Vue.js, and MSSQL; deeply integrated core nutrition management with CRM business logic.
- Real-time Ordering Platform: Developed the platform using Laravel; crafted high-interactivity frontend interfaces and led the architectural design of backend order storage and the "Real-time Settlement & Automated Accounting System."
- **【Traffic or Not】**(Sole Developer): Spearheaded full-stack development using UWP (.NET Framework), driving overall UI design and complex dynamic workflow planning (*Honored with the Global Chinese Sustainable Journalism Award*).
- Virtualization Environment Deployment (Sole Developer): Built and maintained VMware vSphere and Horizon VDI environments, integrating finger-vein biometric authentication to enhance access control; successfully deployed a demo/validation version for the National Chung-Shan Institute of Science and Technology (NCSIST).

(Backend & Data Engineering)

- Yunlin County Government "New Coin" Project: Developed core backend APIs using .NET Core; responsible for rigorous data validation, business logic implementation, and high-performance data restructuring and packaging.
- Antenna Passenger Flow Data Processing: Developed backend APIs using Django (Python) to achieve big data-scale passenger flow data cleaning, multi-dimensional filtering, and anomalous data interception.
- Android Electronic Invoicing App (Sole Developer): Developed a native Android app using Java; dynamically packaged product information and calculated double-barcode formats (1D/QR Code) per Ministry of Finance regulations, achieving deep integration with the government platform.

Trend Micro Taiwan | Network Engineer Assistant

2015.09 - 2019.01

- Developed an automated inspection tool using Python for**【HomeNetworkSecurity】**; resolved network incompatibility pain points and filtered anomalous logs to enhance home cybersecurity defense.
- Conducted IoT vulnerability and threat research; deeply analyzed security flaws in IoT devices and formulated mitigation strategies to effectively identify malicious domains used in social engineering scams.
- Optimized high-concurrency wireless networks; participated in designing and tuning low-latency wireless services, successfully supporting 150,000 concurrent devices and ensuring high availability in high-density environments.

Journal

- Journal IEEE - Improving the Accuracy Rate of PM2.5 Pollution Forecast Using Q-learning training Time-series Model (IEEE INSPEC Accession Number: 19313037)

Conference

- A Tutorial on Using the Sedumi to Solve Optimization Problems Part 1 LP and SCOP Problems (Charng-Kann Chen, CL Chang, 2013 Symposium on Engineering and Technology Applications, 205-208)
- Minimax Design of Two-Dimensional Fan Type FIR Digital Filters Using an Interior-Point Algorithm (Charng-Kann Chen, CL Chang, 2013 Symposium on Engineering and Technology Applications)